

Hypothesis Testing using the GSS

Research topic, independent variable, dependent variable, and quantitative research question.

My topic examines if there is a relationship between one's formal sex education and their perception of their relationship health between an intimate partner or close friends. My independent variable is one's perception of whether sex education should be taught in schools. I use GSS's *sexeduc* variable, which asks if the respondent is in favor of or opposed to sex education in schools. My dependent variable is one's perceived happiness in their relationship with a partner. I use GSS's *hapcohab* variable, which asks if one's relationship with a partner is "very happy," "pretty happy," or "not too happy". My quantitative research question is "do individuals who favor teaching sex education in schools experience happier relationships than those who oppose?"

Research Hypothesis: I expect to find a statistically significant relationship between the variables such that individuals who favor teaching sex education in schools will tend to report having happier relationships with their partners than those who oppose teaching sex education in schools.

Null Hypothesis: There is no statistically significant relationship between one's favor or opposition of sex education in schools and their reported happiness with their relationship.

Data

I am using the 2022 GSS, as my variables are included in several years of the GSS, so the most recent ones may better represent the contemporary context.

The GSS—or General Social Survey—collects data from US adults via personal-interview surveys, and has done so every other year since 1972. It asks respondents for their demographics, attitudes on certain topics, and a range of behaviors. It claims to be the "single best source for sociological and attitudinal trend data covering the United States" and aims to be easily accessible for a range of individuals. Because of the wide range of topics asked by the GSS, data regarding sex education attitudes and relationship happiness is available in most years.

Source: *About the GSS* (<https://gss.norc.umd.edu/en/gss/about-the-gss.html>)

Univariate Analyses

sexeduc asks about one's opinion on having sex-education taught in schools. The question wording is "Would you be for or against sex education in the public schools?" The valid attributes are nominal and include the following:

1. Favor
2. Oppose

The invalid attributes include the following:

- 100. Inapplicable
- 99. No answer
- 98. Do not Know/Cannot Choose
- 97. Skipped on Web

hapcohab asks about the respondent's happiness in their relationship with their partner. The question wording is "Taking things all together, would you say that your relationship with your partner is very happy, pretty happy, or not too happy?" The valid attributes are ordinal, and they include the following:

1. Very happy
2. Pretty happy
3. Not too happy

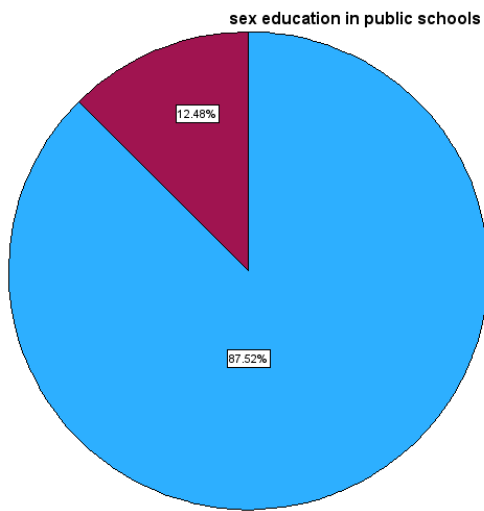
The invalid attributes include the following:

- 100. Inapplicable
- 99. No answer
- 98. Do not Know/Cannot Choose
- 97. Skipped on Web

Frequency distributions and measures of central tendency and dispersion

sex education in public schools						Statistics		
		Frequency	Percent	Valid Percent	Cumulative Percent	sex education in public schools		
Valid	favor	2384	57.5	87.5	87.5	N	Valid	2724
	oppose	340	8.2	12.5	100.0		Missing	1425
	Total	2724	65.7	100.0		Mode		
Missing	System	1425	34.3			Minimum		
Total		4149	100.0			Maximum		

happiness of relt with partner						Statistics		
		Frequency	Percent	Valid Percent	Cumulative Percent	happiness of relt with partner		
Valid	very happy	211	5.1	53.4	53.4	N	Valid	395
	pretty happy	161	3.9	40.8	94.2		Missing	3754
	not too happy	23	.6	5.8	100.0	Median		
	Total	395	9.5	100.0		Mode		
Missing	System	3754	90.5			Minimum		
Total		4149	100.0			Maximum		



sexeduc:

The maximum value is 2 (oppose) and the minimum value is 1 (favor). The only measure of central tendency for *sexeduc* is the mode (as this is a nominal variable), which is 1 (favor). The dispersion of the variable shows that a vast majority of the valid responses (87.5%) favored sex education in school, with the other 12.5% opposing. This is also reflected in the pie chart, which shows the same percentages and visualizes the extent to which respondents favored option 1.

hapcohab:

The maximum value is 3 (“not too happy”) and the minimum value is 1 (“very happy”). The measures of central tendency appropriate for this variable are mode and median, as this is an ordinal variable. The mode and median are both 1 (very happy) with over half (53.4%) of the respondents selecting this option. The distribution of the data suggests that it is right-skewed, with just 5.8% selecting the maximum value 3 (“not too happy”). This distribution is also apparent with the bar chart, which shows the highest-chosen value to be “very happy” followed by “pretty happy.”

Bivariate Hypothesis Testing Case Processing Summary

	Valid		Missing		Total	
	N	Percent	N	Percent	N	Percent
happiness of relt with partner * sex education in public schools	263	6.3%	3886	93.7%	4149	100.0%

happiness of relt with partner * sex education in public schools Crosstabulation

			sex education in public schools		
			favor	oppose	Total
happiness of relt with partner	very happy	Count	127	12	139
		% within sex education in public schools	53.6%	46.2%	52.9%
	pretty happy	Count	99	9	108
		% within sex education in public schools	41.8%	34.6%	41.1%
	not too happy	Count	11	5	16
		% within sex education in public schools	4.6%	19.2%	6.1%
Total		Count	237	26	263
		% within sex education in public schools	100.0%	100.0%	100.0%

The above crosstab reflects that those opposed to sex education in schools had proportionately higher reports of being “not too happy” with their relationship than their favoring counterparts. Just 4.6% of individuals who favored sex education in schools reported being “not too happy,” compared to the 19.2% of those opposed to sex education who also reported being “not too happy” with their relationship. A higher percentage of individuals favoring sex education reported being “very happy” (53.6%) with their relationship than those who opposed (46.2%).

Test of Significance and Measure of Association analyses.

Chi-Square Tests

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	8.735 ^a	2	.013
Likelihood Ratio	6.118	2	.047
Linear-by-Linear Association	3.054	1	.081
N of Valid Cases	263		

a. 1 cells (16.7%) have expected count less than 5. The minimum expected count is 1.58.

Symmetric Measures

		Value	Approximate Significance
Nominal by Nominal	Phi	.182	.013
	Cramer's V	.182	.013
N of Valid Cases		263	

The Chi-Square test of significance found a p-value of 0.013. Lambda was giving me an error, so I used Phi for the measure of association. This value was 0.182.

I planned to use Lambda as my measure of association, as this is also appropriate for nominal variables. Unfortunately lambda was not working, so I chose to use Phi as my measure of association, at the recommendation of Professor Parris.

Discussion

The test of significance found a p-value of 0.013. This is under the 0.05 threshold for statistical significance, so I can conclude that a statistically significant relationship exists between how one favors

or opposes sex education in schools and their perceived happiness with a partner. In other words, I can reject the null hypothesis that there is *not* a statistically significant relationship between the variables.

The measure of association found a value of 0.182. This value is positive, indicating that there is a positive correlation between the variables. The way that these variables are coded, this means that individuals who oppose sex education (i.e. those who responded 2 instead of 1) are more associated with reporting higher values for *hapcohab* as well (i.e. less happiness with their partner). The measure of association's value of 0.182 is closer to 0 than 1. Typically, values from lambda that are less than 0.2 are considered weakly correlated. This might suggest that while there is a statistically significant correlation between the variables, one's views on sex education in schools are not considered to be the main factor contributing to their happiness with a partner.

We might begin to explain this weak correlation by considering one's experiences with sex education, and how this might be another variable to consider. One might report favoring sex education in schools if they valued their own sex education. Having a good sex education (which is possibly indicated by wanting sex education to continue) might have given respondents experiences and values that allowed them to pursue happy relationships. Another possibility is that individuals with negative feelings about sex education in schools may have poor experiences with sex and relationships, and are more likely to report unhappiness with a partner. It is important to consider in our analysis that this correlation between valuing/opposing sex education in schools and relationship happiness is weak, and that there may be many more variables involved in one's degree of relationship happiness.